

Advanced Probabilistic Machine Learning

Approximate Inference: Variational Inference

Ralf Herbrich, Rainer Schlosser

Overview

1. Minimizing Reverse KL-Divergence via ELBO
2. Variational Inference for Gaussian Factors
3. Application: Robust Bayesian Regression and Ranking
4. Mixture Models

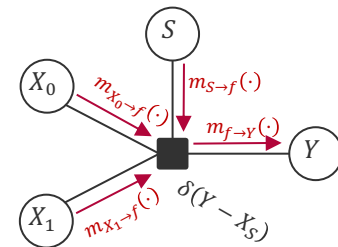
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*Unit 6 – Approximate Inference:
Variational Inference*

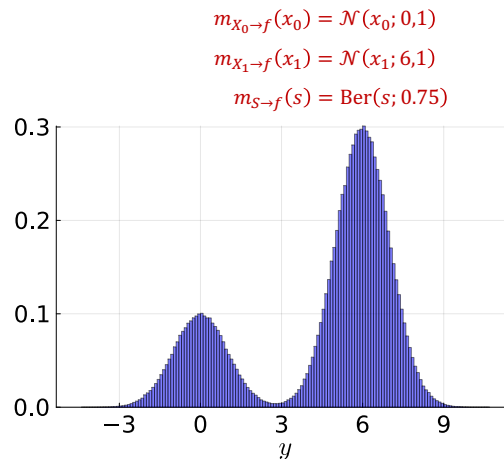
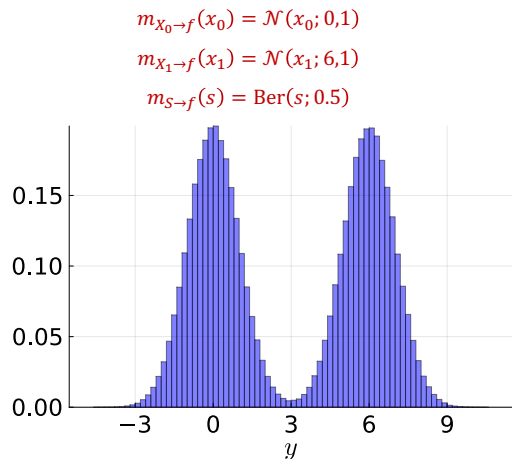
Motivation: Mixture Models

- **Mixture Model:** If we have two possible causes X_0 and X_1 for an observed effect Y and a (non-observed) binary selector $S \in \{0,1\}$ then

$$f(X_0, X_1, S, Y) = \sum_{s=0}^1 \mathbb{I}(S = s) \cdot \delta(Y - X_s) = \delta(Y - X_S)$$



- **Example:** A match outcome model where the skill of a player is a mixture of the skills of two real people in a household



```
# implements the mixture factor
function mixture(x0, x1, s)
    return (s == 0) ? x0 : x1
end

# samples the message to the output variable for a mixture factor
function sample_mixture_message()
    mu_x0 = 0.0, sigma_x0=1.0, mu_x1=6.0, sigma_x1=1.0, p=0.5,
    n_samples=1000000
)

# stores the samples of y
samples_msg_to_y = Vector{Float64}{}(undef, n_samples)
for i in 1:n_samples
    x0 = randn() * sigma_x0 + mu_x0
    x1 = randn() * sigma_x1 + mu_x1
    s = rand() < p ? 1 : 0
    # compute y using the mixture factor
    samples_msg_to_y[i] = mixture(x0, x1, s)
end
end
```

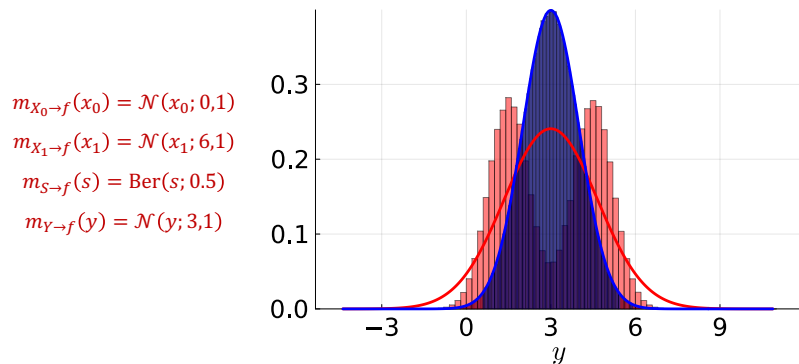
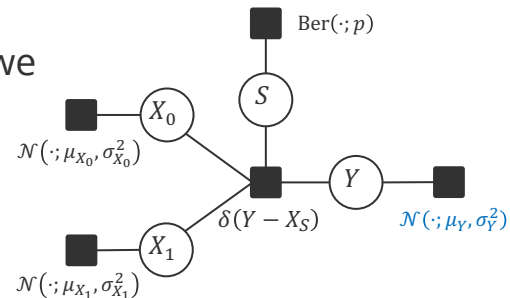
Motivation: Mixture Models (ctd)

- **Expectation Propagation:** Since the message $m_{f \rightarrow Y}(\cdot)$ is not a Gaussian, we approximate the marginal $p_Y(\cdot) = m_{f \rightarrow Y}(\cdot) \cdot m_{Y \rightarrow f}(\cdot)$ with $\hat{p}_Y(\cdot)$ by virtue of

$$\hat{p}_Y(\cdot) = \mathcal{N}(\cdot; \underbrace{E_{Y \sim p_Y}[Y]}_{\hat{\mu}_Y}, \underbrace{\text{var}_{Y \sim p_Y}[Y]}_{\hat{\sigma}_Y^2})$$

- **Idea:** Instead of algebraic integration, use importance sampling with the proposal

$$q(x_0, x_1, s, y) = \mathcal{N}(x_0; \mu_0, \sigma_0^2) \cdot \mathcal{N}(x_1; \mu_1, \sigma_1^2) \cdot \text{Ber}(s; p) \cdot \delta(y - x_s) \quad \Rightarrow \quad w(x_0, x_1, s, y) = \frac{\prod_f f(x_0, x_1, s, y)}{q(x_0, x_1, s, y)} = \mathcal{N}(y; \mu_Y, \sigma_Y^2)$$



$$\hat{m}_{f \rightarrow Y}(y) = \frac{\hat{p}_Y(y)}{m_{Y \rightarrow f}(y)}$$

$$= \frac{\mathcal{N}(y; \hat{\mu}_Y, \hat{\sigma}_Y^2)}{\mathcal{N}(y; \mu_Y, \sigma_Y^2)}$$

Not possible because $\hat{\sigma}_Y^2 > \sigma_Y^2$

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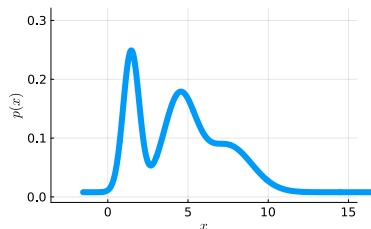
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Recall: α -Divergence

- **α -Divergence (Amari, 1985).** Given two probability densities $p(\cdot)$ and $q(\cdot)$ and $\alpha \in \mathbb{R} \setminus \{0,1\}$ the α -divergence $D_\alpha[p, q]$ is defined by

$$D_\alpha[p, q] = \frac{1}{\alpha(1-\alpha)} \cdot \left(1 - \int_{-\infty}^{+\infty} \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) dx \right)$$



zero-forcing & mode seeking

inclusive & support seeking

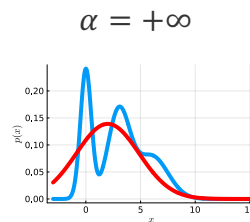
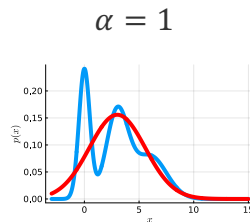
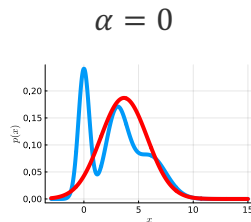
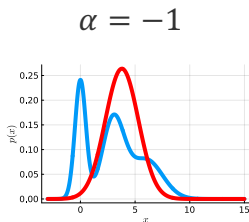
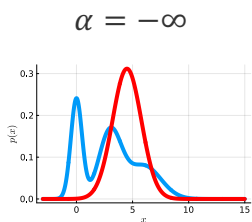
α



Shun'ichi Amari (甘利 俊)
(1936)

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$\alpha = 0$: Reverse KL Divergence

- **Theorem (Limit $\alpha \rightarrow 0$).** Given two probability densities $p(\cdot)$ and $q(\cdot)$ the limit of the α -divergence $D_\alpha[p, q]$ for $\alpha \rightarrow 0$ is the reverse Kullback-Leibler divergence

Note that for $\alpha \rightarrow 1$ we get $\text{KL}[p(\cdot), q(\cdot)]$

$$\lim_{\alpha \rightarrow 0} D_\alpha[p, q] \Rightarrow \text{KL}[q(\cdot), p(\cdot)] := \int_{-\infty}^{+\infty} \log\left(\frac{q(x)}{p(x)}\right) \cdot q(x) \, dx$$

- **Proof:** Taking limits, we have

$$\begin{aligned} \lim_{\alpha \rightarrow 0} D_\alpha[p, q] &= \lim_{\alpha \rightarrow 0} \frac{1}{\alpha(1-\alpha)} \cdot \left(1 - \int_{-\infty}^{+\infty} \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right) \quad \text{L'Hôpital's rule!} \\ &= \lim_{\alpha \rightarrow 0} \frac{1}{1-2\alpha} \cdot \left(- \int_{-\infty}^{+\infty} \log\left(\frac{p(x)}{q(x)}\right) \cdot \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right) \quad \text{Note that } \frac{d}{dz} b^z = \log(b) \cdot b^z \\ &= \int_{-\infty}^{+\infty} \log\left(\frac{q(x)}{p(x)}\right) \cdot q(x) \, dx \end{aligned}$$

- **Challenge:** Given an arbitrary density $p(\cdot)$ how do we efficiently find the minimizer of the reverse Kullback-Leibler divergence when

$$q(\cdot) = \exp\left(\boldsymbol{\theta}^T \mathbf{T}(\cdot) - A(\boldsymbol{\theta})\right) \quad \text{Exponential family parameterized by } \mathbf{T}(\cdot)$$



Solomon Kullback
(1909 – 1994)



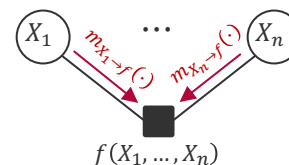
Richard Leibler
(1914 – 2003)

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Evidence Lower Bound (ELBO)

- Consider a single factor f of n non-observed variables $\mathbf{X} := X_1, \dots, X_n$
 - All other non-observed variables $X_{n+1}, X_{n+2}, \dots$ have already been integrated out
 - All observed variables (i.e., data D) are encoded through other factors and f



$$p(\mathbf{x}, D) = f(\mathbf{x}_1, \dots, \mathbf{x}_n) \cdot \prod_{i=1}^n m_{X_i \rightarrow f(\cdot)}(x_i)$$

- **Observation 1:** Since the data D is observed, we know that $p(D)$ is constant
- **Observation 2:** For *any* distribution q over $X_1, \dots, X_n$ we know that

$$\begin{aligned} \log(p(D)) &= \log(p(D)) \cdot \int q(\mathbf{x}) \, d\mathbf{x} \quad \leftarrow = 1 \\ &= \int q(\mathbf{x}) \cdot \left[\log\left(\frac{p(\mathbf{x}, D)}{p(\mathbf{x}|D)} \cdot \frac{q(\mathbf{x})}{q(\mathbf{x})}\right) \right] d\mathbf{x} \quad \leftarrow = 1 \\ &= \int q(\mathbf{x}) \cdot \left[\log\left(\frac{p(\mathbf{x}, D)}{q(\mathbf{x})}\right) \right] d\mathbf{x} + \int q(\mathbf{x}) \cdot \left[\log\left(\frac{q(\mathbf{x})}{p(\mathbf{x}|D)}\right) \right] d\mathbf{x} \end{aligned}$$

ELBO[$q(\cdot), p(\cdot, D)$]

KL[$q(\cdot), p(\cdot | D)$] ≥ 0

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Evidence Lower Bound (ELBO): Example

$$\text{KL}[q(\cdot), p(\cdot | D)] = \log(p(D)) - \text{ELBO}[q(\cdot), p(\cdot, D)] \geq 0$$

$$\log(p(D)) \geq \text{ELBO}[q(\cdot), p(\cdot, D)]$$

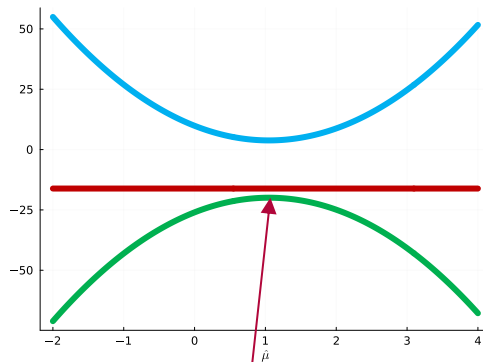
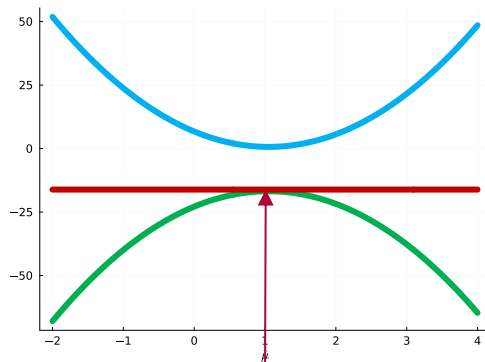
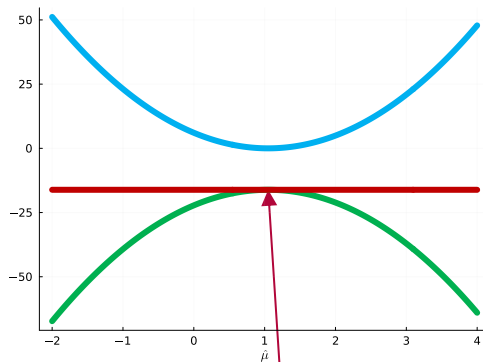
$\text{ELBO}[q(\cdot), p(\cdot, D)]$ is a lower bound of the (constant) evidence $\log(p(D))$

$$p(X, D) = \mathcal{N}(X; 0, 1) \cdot \prod_{i=1}^{10} \mathcal{N}(x_i; X, 1)$$

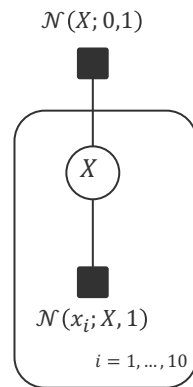
$$q(\cdot) = \mathcal{N}\left(\cdot; \hat{\mu}, \frac{1}{11}\right)$$

$$q(\cdot) = \mathcal{N}\left(\cdot; \hat{\mu}, \frac{1}{100}\right)$$

$$q(\cdot) = \mathcal{N}(\cdot; \hat{\mu}, 1)$$



Maximizing $\text{ELBO}[q(\cdot), p(\cdot, D)]$ over $q(\cdot)$ finds the minimizer of $\text{KL}[q(\cdot), p(\cdot | D)]$



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ELBO for Factorizing Distributions

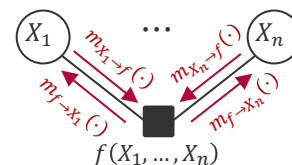
- **Assumption:** Since we are only interested in the marginals $p_{X_i}(\cdot)$ we will assume

$$q(x_1, \dots, x_n) = \prod_{i=1}^n q_i(x_i)$$

- For our factor graph model, we have $q_i(x_i) \propto m_{f \rightarrow X_i}(x_i) \cdot m_{X_i \rightarrow f}(x_i)$

- In this case the ELBO function becomes a sum of two terms

$$\begin{aligned} \text{ELBO}[q(\cdot), p(\cdot, D)] &= \int q(\mathbf{x}) \cdot \left[\log \left(\frac{p(\mathbf{x}, D)}{q(\mathbf{x})} \right) \right] d\mathbf{x} \\ &= \int \left[\prod_{i=1}^n q_i(x_i) \right] \cdot \left[\log \left(\frac{p(\mathbf{x}, D)}{\prod_{i=1}^n q_i(x_i)} \right) \right] d\mathbf{x} \\ &= \int \left[\prod_{i=1}^n q_i(x_i) \right] \cdot \log(p(\mathbf{x}, D)) d\mathbf{x} - \sum_{i=1}^n \int q_i(x_i) \cdot \log(q_i(x_i)) dx_i \end{aligned}$$



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Maximizing ELBO for Factorizing Distributions

- **Idea:** Assuming only $q_j(\cdot)$ is unknown, we derive the maximizer of ELBO.

$$\begin{aligned}
 \text{ELBO}[q_j(\cdot), p(\cdot, D)] &= \int \left[\prod_{i=1}^n q_i(x_i) \right] \cdot \log(p(\mathbf{x}, D)) \, d\mathbf{x} - \sum_{i=1}^n \int q_i(x_i) \cdot \log(q_i(x_i)) \, dx_i \\
 &= \int q_j(x_j) \cdot \left(\int \left[\prod_{i \neq j}^n q_i(x_i) \right] \cdot \log(p(\mathbf{x}, D)) \, d\mathbf{x}_{\setminus j} \right) dx_j - \int q_j(x_j) \cdot \log(q_j(x_j)) \, dx_j + \text{const} \\
 &= - \int q_j(x_j) \cdot \log \left(\frac{q_j(x_j)}{\exp(E_j(x_j))} \right) dx_j + \text{const}
 \end{aligned}$$

$q(x_1, \dots, x_n) = q_j(x_j) \cdot \prod_{i \neq j}^n q_i(x_i)$
 (The product term is labeled "const" with an arrow pointing to it.)

$E_j(x_j)$ where all X_i with $i \neq j$ are integrated out

$\text{KL}[q(\cdot), \exp(E_j(\cdot))] \geq 0$ with equality if and only if $q(\cdot) \propto \exp(E_j(\cdot))$

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- Thus, the ELBO maximiser $q_j^*(\cdot)$ must fulfill the stationary condition:

$$q_j^*(x_j) \propto \exp(E_j(x_j))$$

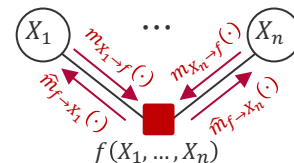
Note that the $E_j(\cdot)$ requires all other $q_i(\cdot)$ so these are n coupled conditions for the ELBO maximiser!

Variational Inference for Exponential Family

- Consider a single factor f of n non-observed variables $\mathbf{X} := X_1, \dots, X_n$

$$p(\mathbf{x}, D) = f(x_1, \dots, x_n) \cdot \prod_{i=1}^n m_{X_i \rightarrow f}(x_i)$$

- Assumption 1:** All incoming messages from the variables X_i are in the exponential family parameterized by the sufficient statistic $\mathbf{T}_i(\cdot)$



$$m_{X_i \rightarrow f}(x_i) \propto \exp(\boldsymbol{\theta}_i^T \mathbf{T}_i(x_i))$$

- Assumption 2:** The factor f has the n properties that for all $i = 1, \dots, n$

$$f(x_1, \dots, x_n) \propto \exp(\mathbf{O}_i(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)^T \mathbf{T}_i(x_i))$$

$$q_i^*(x_i) \propto \exp\left(\int \left[\prod_{j \neq i}^n q_j(x_j)\right] \cdot \log(p(\mathbf{x}, D)) \, d\mathbf{x}_{\setminus i}\right)$$

- Then, the variational approximation $q_i^*(\cdot)$ of the marginal $p_{X_i}(\cdot)$ is

$$q_i^*(x_i) \propto \exp\left(\boldsymbol{\theta}_i^T \mathbf{T}_i(x_i) + E_{X_1 \sim q_1^*, \dots, X_{i-1} \sim q_{i-1}^*, X_{i+1} \sim q_{i+1}^*, \dots, X_n \sim q_n^*}[\mathbf{O}_i(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n)]^T \mathbf{T}_i(x_i)\right)$$

- Dividing out $m_{X_i \rightarrow f}(x_i)$, the variational approximation $\hat{m}_{f \rightarrow X_i}(\cdot)$ is given by

$$\hat{m}_{f \rightarrow X_i}(x_i) \propto \exp\left(E_{X_1 \sim q_1^*, \dots, X_{i-1} \sim q_{i-1}^*, X_{i+1} \sim q_{i+1}^*, \dots, X_n \sim q_n^*}[\mathbf{O}_i(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n)]^T \mathbf{T}_i(x_i)\right)$$

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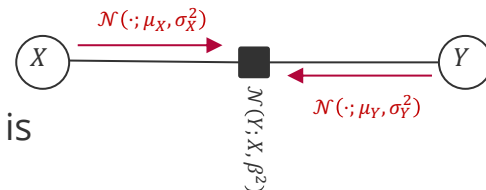
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Gaussian Mean Factor: Joint Distribution

$$f(X, Y) = \mathcal{N}(Y; X, \beta^2)$$



- The true **joint distribution** given this factor and the incoming messages is

$$p(x, y) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \mathcal{N}(y; \mu_Y, \sigma_Y^2) \cdot \mathcal{N}(y; x, \beta^2)$$

$$\propto \mathcal{N} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} \mu_X \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \\ \mu_X \cdot \frac{\sigma_Y^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \end{bmatrix}, \begin{bmatrix} \sigma_X^2 \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} & \sigma_X^2 \sigma_Y^2 \cdot \frac{1}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \\ \sigma_X^2 \sigma_Y^2 \cdot \frac{1}{\sigma_X^2 + \sigma_Y^2 + \beta^2} & \sigma_Y^2 \cdot \frac{\sigma_X^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \end{bmatrix} \right)$$

Weighted average of the means with weights proportional to the variance ratios

Scaled (co)variances with weights proportional to the variance ratios

- Derived using the natural parameterization and collecting terms for x , y , x^2 , y^2 and $x \cdot y$
- In the limit case of $\beta \rightarrow \infty$, the factor $\mathcal{N}(y; x, \beta^2)$ becomes the uniform-Normal and

$$p(x, y) \propto \mathcal{N} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \sigma_X^2 & 0 \\ 0 & \sigma_Y^2 \end{bmatrix} \right)$$

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Gaussian Mean Factor: Variational Approximation

$$\begin{aligned}
 f(X, Y) &= \mathcal{N}(Y; X, \beta^2) \\
 &= \mathcal{G}(Y; \beta^{-2}X, \beta^{-2}) \propto \exp \left(\begin{bmatrix} \beta^{-2}Y \\ \beta^{-2} \end{bmatrix}^T \begin{bmatrix} X \\ X^2 \\ -\frac{1}{2} \end{bmatrix} \right) \\
 &= \mathcal{G}(X; \beta^{-2}Y, \beta^{-2}) \propto \exp \left(\begin{bmatrix} \beta^{-2}X \\ \beta^{-2} \end{bmatrix}^T \begin{bmatrix} Y \\ Y^2 \\ -\frac{1}{2} \end{bmatrix} \right)
 \end{aligned}$$

Diagram illustrating the Gaussian Mean Factor Variational Approximation. The central node is a red square representing the joint distribution $\mathcal{N}(Y; X, \beta^2)$. It is connected to node X (circle) and node Y (circle). The forward message from X to the square is $\mathcal{N}(\cdot; \mu_X, \sigma_X^2)$, and the backward message is $\hat{m}_{f \rightarrow X}(\cdot)$. The forward message from Y to the square is $\hat{m}_{f \rightarrow Y}(\cdot)$, and the backward message is $\mathcal{N}(\cdot; \mu_Y, \sigma_Y^2)$. The square is also labeled with $\mathcal{N}(Y; X, \beta^2)$. The equations for the variational approximations are:

$$\begin{aligned}
 \hat{m}_{f \rightarrow X}(x) &\propto \exp \left(E_{Y \sim \hat{p}_Y} [\mathbf{O}_X(Y)]^T \mathbf{T}_X(x) \right) \propto \mathcal{G}(x; \beta^{-2} E_{Y \sim \hat{p}_Y} [Y], \beta^{-2}) \\
 \hat{m}_{f \rightarrow Y}(y) &\propto \exp \left(E_{X \sim \hat{p}_X} [\mathbf{O}_Y(X)]^T \mathbf{T}_Y(y) \right) \propto \mathcal{G}(y; \beta^{-2} E_{X \sim \hat{p}_X} [X], \beta^{-2})
 \end{aligned}$$

Side equations for the joint distributions:

$$\begin{aligned}
 \hat{p}_Y(y) &= \mathcal{N}(y; \mu_Y, \sigma_Y^2) \cdot \hat{m}_{f \rightarrow Y}(y) \\
 \hat{p}_X(x) &= \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \hat{m}_{f \rightarrow X}(x)
 \end{aligned}$$

- The two messages are coupled due to the expectation of X and Y !
- We can analytically solve these two equations for the joint solution to obtain

$$\hat{p}(x, y) \propto \mathcal{N} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} \mu_X \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \\ \mu_X \cdot \frac{\sigma_Y^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} + \mu_Y \cdot \frac{\sigma_X^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2} \end{bmatrix}, \begin{bmatrix} \sigma_X^2 \cdot \frac{\beta^2}{\sigma_X^2 + \beta^2} & 0 \\ 0 & \sigma_Y^2 \cdot \frac{\beta^2}{\sigma_Y^2 + \beta^2} \end{bmatrix} \right)$$

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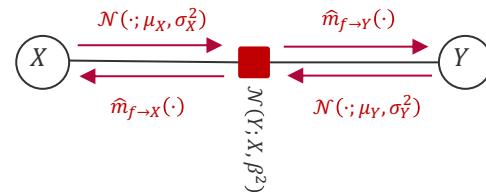
Smaller than the true variance of $\sigma_X^2 \cdot \frac{\sigma_Y^2 + \beta^2}{\sigma_X^2 + \sigma_Y^2 + \beta^2}$

Gaussian Mean Factor: Examples

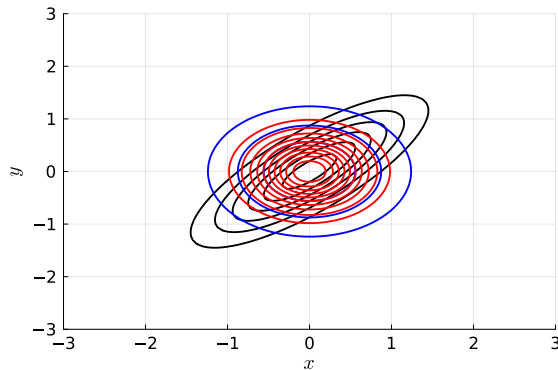
$$m_{X \rightarrow f}(x) = \mathcal{N}(x; 0, 1)$$

$$m_{Y \rightarrow f}(y) = \mathcal{N}(y; 0, 1)$$

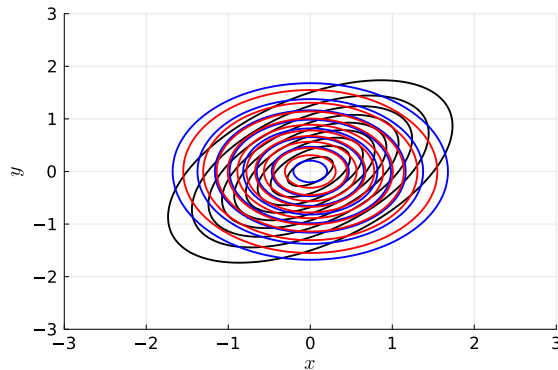
$$f(x, y) = \mathcal{N}(y; x, \beta^2)$$



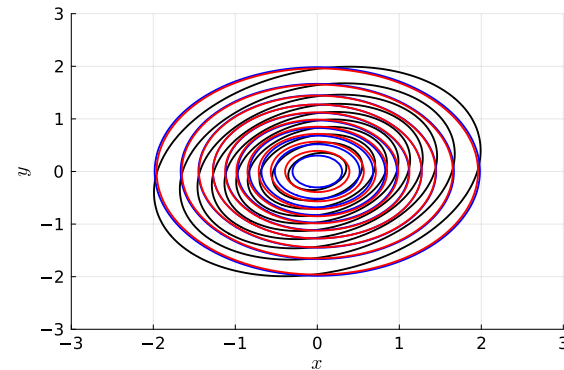
$\beta = 0.5$



$\beta = 1$



$\beta = 2$



Gaussian Factor with Variable Precision

$$f(X, S) = \mathcal{N}(X; \mu, S^{-1})$$

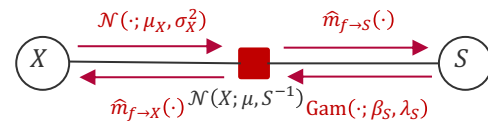
$$= \exp\left(\frac{1}{2}\log(S) - \frac{1}{2}\log(2\pi) - \frac{S \cdot (X - \mu)^2}{2}\right)$$

$$\propto \exp\left(\begin{bmatrix} S \cdot \mu \\ S \end{bmatrix}^T \begin{bmatrix} X \\ X^2 \\ -\frac{1}{2} \end{bmatrix}\right)$$

$$\propto \exp\left(\begin{bmatrix} \frac{1}{2} \\ \frac{(X - \mu)^2}{2} \\ \log(S) \\ -S \end{bmatrix}^T \begin{bmatrix} \log(S) \\ -S \end{bmatrix}\right)$$

$$\hat{m}_{f \rightarrow X}(x) \propto \exp\left(E_{S \sim \hat{p}_S}[\mathbf{O}_X(S)]^T \mathbf{T}_X(x)\right) \\ \propto \mathcal{G}(x; \mu, E_{S \sim \hat{p}_S}[S], E_{S \sim \hat{p}_S}[S])$$

$$\hat{m}_{f \rightarrow S}(s) \propto \exp\left(E_{X \sim \hat{p}_X}[\mathbf{O}_S(X)]^T \mathbf{T}_S(s)\right) \\ \propto \text{Gam}\left(s; \frac{1}{2}, \frac{E_{X \sim \hat{p}_X}[X^2] + \mu^2}{2} - \mu \cdot E_{X \sim \hat{p}_X}[X]\right)$$



$$\hat{p}_S(s) = \text{Gam}(s; \beta_S, \lambda_S) \cdot \hat{m}_{f \rightarrow S}(s)$$

$$\hat{p}_X(x) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \hat{m}_{f \rightarrow X}(x)$$

Advanced Probabilistic
Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

- The two messages are coupled due to the expectation of X and S !
- There is no analytical solution to solve them simultaneously
- Thus, they get iterated until convergence from a start-point given by the incoming messages $\mathcal{N}(\cdot; \mu_X, \sigma_X^2)$ and $\text{Gam}(\cdot; \beta_S, \lambda_S)$

Gaussian Factor with Variable Precision: Examples

$$m_{X \rightarrow f}(x) = \mathcal{N}(x; 0, 1)$$

$$m_{S \rightarrow f}(s) = \text{Gam}(s; 3, 0.5)$$

$$f(x, s) = \mathcal{N}(x; 2, s^{-1})$$

```
function variational_message_passing_gaussian_factor(msg_from_X_to_f::Gaussian1D, msg_from_S_to_f::Gamma1D, y)
    msg_from_f_to_X = Gaussian.Gaussian1DUniform()
    msg_from_f_to_S = Gamma.Gamma1DUniform()

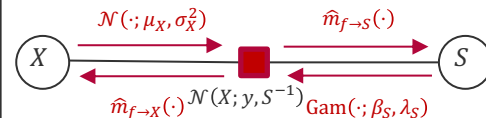
    while true
        # cache the old marginals
        p_X_old = msg_from_f_to_X * msg_from_X_to_f
        p_S_old = msg_from_f_to_S * msg_from_S_to_f

        # obtain the current marginal of S and compute the variational approximation of the message from f to X
        p_S = msg_from_f_to_S * msg_from_S_to_f
        msg_from_f_to_X = Gaussian.Gaussian1D(y * Gamma.mean(p_S), Gamma.mean(p_S))

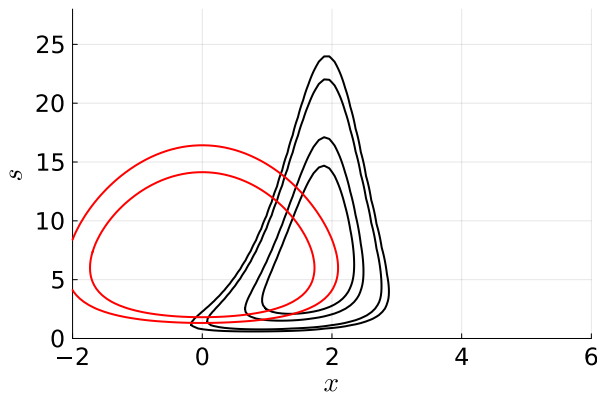
        # Update message from factor f to X
        p_X = msg_from_f_to_X * msg_from_X_to_f
        msg_from_f_to_S = Gamma.Gamma1D(0.5, 1/2 * (y^2 + Gaussian.variance(p_X) + (Gaussian.mean(p_X))^2) - y * Gaussian.mean(p_X))

        # compute the new marginals
        p_X_new = msg_from_f_to_X * msg_from_X_to_f
        p_S_new = msg_from_f_to_S * msg_from_S_to_f

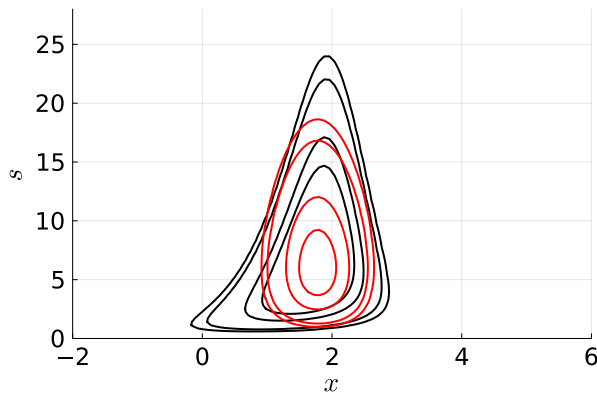
        if Gaussian.KL_divergence(p_X_old, p_X_new) < 1e-6 && Gamma.KL_divergence(p_S_old, p_S_new) < 1e-6
            return (p_X_new, p_S_new)
        end
    end
end
```



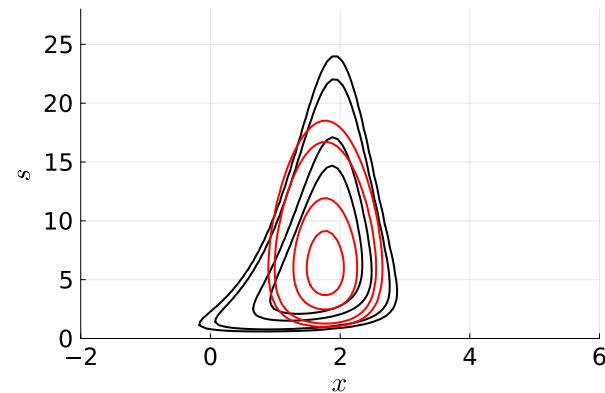
Start



Iteration 1



Iteration 2



Gaussian Mean Factor with Variable Precision

$$f(X, Y, S) = \mathcal{N}(X; Y, S^{-1}) = \exp\left(\frac{1}{2}\log(S) - \frac{1}{2}\log(2\pi) - \frac{S \cdot (X - Y)^2}{2}\right)$$

$$\begin{aligned} \mathbf{o}_X(S, Y) &\propto \exp\left(\begin{bmatrix} S \cdot Y \\ S \end{bmatrix}^T \begin{bmatrix} X \\ -\frac{X^2}{2} \end{bmatrix}\right) \quad \mathbf{T}_X(X) \end{aligned}$$

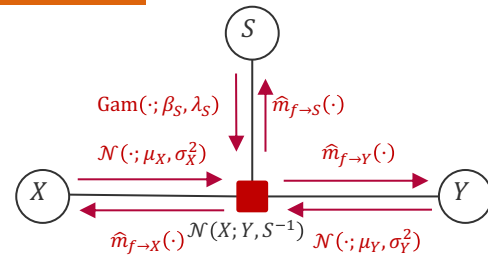
$$\mathbf{o}_Y(S, X) \propto \exp\left(\begin{bmatrix} S \cdot X \\ S \end{bmatrix}^T \begin{bmatrix} Y \\ -\frac{Y^2}{2} \end{bmatrix}\right) \quad \mathbf{T}_Y(Y)$$

$$\mathbf{o}_S(X, Y) \propto \exp\left(\begin{bmatrix} \frac{1}{2} \\ \frac{(X - Y)^2}{2} \\ -S \end{bmatrix}^T \begin{bmatrix} \log(S) \\ -S \end{bmatrix}\right) \quad \mathbf{T}_S(S)$$

$$\begin{aligned} \hat{m}_{f \rightarrow X}(x) &\propto \exp\left(E_{Y \sim \hat{p}_Y, S \sim \hat{p}_S}[\mathbf{o}_X(S, Y)]^T \mathbf{T}_X(x)\right) \\ &\propto \mathcal{G}(x; E_{Y \sim \hat{p}_Y}[Y] \cdot E_{S \sim \hat{p}_S}[S], E_{S \sim \hat{p}_S}[S]) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow Y}(y) &\propto \exp\left(E_{X \sim \hat{p}_X, S \sim \hat{p}_S}[\mathbf{o}_Y(S, X)]^T \mathbf{T}_Y(y)\right) \\ &\propto \mathcal{G}(y; E_{X \sim \hat{p}_X}[X] \cdot E_{S \sim \hat{p}_S}[S], E_{S \sim \hat{p}_S}[S]) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow S}(s) &\propto \exp\left(E_{X \sim \hat{p}_X, Y \sim \hat{p}_Y}[\mathbf{o}_S(X, Y)]^T \mathbf{T}_S(s)\right) \\ &\propto \text{Gam}\left(s; \frac{1}{2}, \frac{E_{X \sim \hat{p}_X}[X^2] + E_{Y \sim \hat{p}_Y}[Y^2]}{2} - E_{Y \sim \hat{p}_Y}[Y] \cdot E_{X \sim \hat{p}_X}[X]\right) \end{aligned}$$



$$\hat{p}_S(s) = \text{Gam}(s; \beta_S, \lambda_S) \cdot \hat{m}_{f \rightarrow S}(s)$$

$$\hat{p}_Y(y) = \mathcal{N}(y; \mu_Y, \sigma_Y^2) \cdot \hat{m}_{f \rightarrow Y}(y)$$

$$\hat{p}_X(x) = \mathcal{N}(x; \mu_X, \sigma_X^2) \cdot \hat{m}_{f \rightarrow X}(x)$$

Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

- The three messages (six equations) are coupled due to the expectation of X , Y and S
- Iterate to solve for the six parameters of the messages $\hat{m}_{f \rightarrow X}(\cdot)$, $\hat{m}_{f \rightarrow Y}(\cdot)$ and $\hat{m}_{f \rightarrow S}(\cdot)$

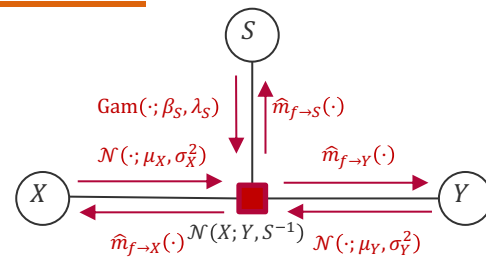
Gaussian Factor with Variable Precision: Examples

$$m_{X \rightarrow f}(x) = \mathcal{N}(x; 0, 1)$$

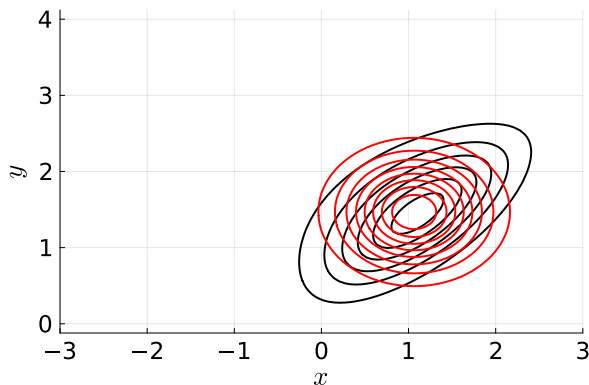
$$m_{Y \rightarrow f}(y) = \mathcal{N}(y; 2, 0.5)$$

$$m_{S \rightarrow f}(s) = \text{Gam}(s; 2, 1)$$

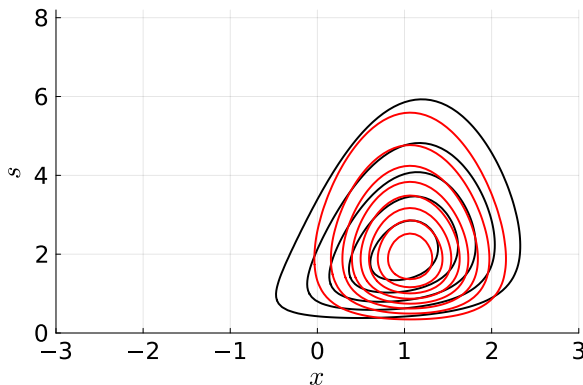
$$f(x, y, s) = \mathcal{N}(x; y, s^{-1})$$



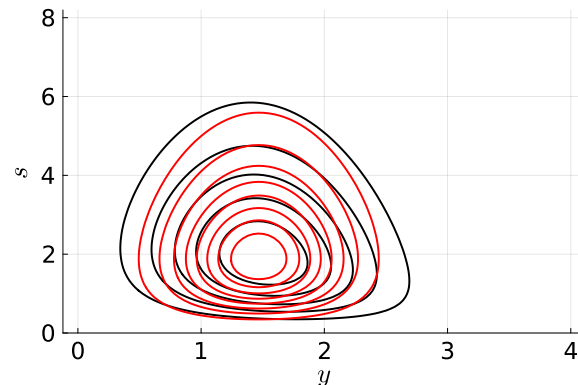
Marginal p_{XY}



Marginal p_{XS}



Marginal p_{YS}



Overview

1. Minimizing Reverse KL-Divergence via ELBO
2. Variational Inference for Gaussian Factors
- 3. Application: Robust Bayesian Regression and Ranking**
4. Mixture Models

**Advanced Probabilistic
Machine Learning**

*Unit 6 – Approximate Inference:
Variational Inference*

Robust Bayesian Regression

■ Given:

1. **Training Data:** $D \in (\mathcal{X} \times \mathbb{R})^n$ of n (labelled) examples (x_i, y_i)
2. **Linear Basis Functions:** Basis function mapping $\phi: \mathcal{X} \rightarrow \mathbb{R}^M$ and linear function model $f(x; \mathbf{w}):= \mathbf{w}^T \phi(x)$
3. **Per-Data Point Precisions** $s_i \in \mathbb{R}^+$
4. **Likelihood of functions and precisions:**

$$p(D|\mathbf{w}, \mathbf{s}) = \prod_{i=1}^n \mathcal{N}(y_i; \mathbf{w}^T \phi(x_i), s_i^{-1})$$

5. **Prior belief over functions and precisions:**

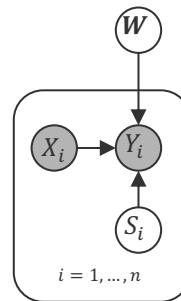
$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$p(\mathbf{s}) = \prod_{i=1}^n \text{Gam}(s_i; \beta, \lambda)$$

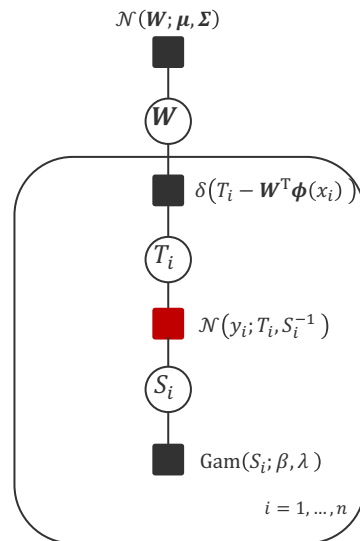
- ## ■ Bayesian Inference:
- Posterior belief over functions and precisions

$$p(\mathbf{w}, \mathbf{s}|D) \propto \left[\prod_{i=1}^n \mathcal{N}(y_i; \mathbf{w}^T \phi(x_i), s_i^{-1}) \cdot \text{Gam}(s_i; \beta, \lambda) \right] \cdot \mathcal{N}(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Bayesian Network



Factor Graph

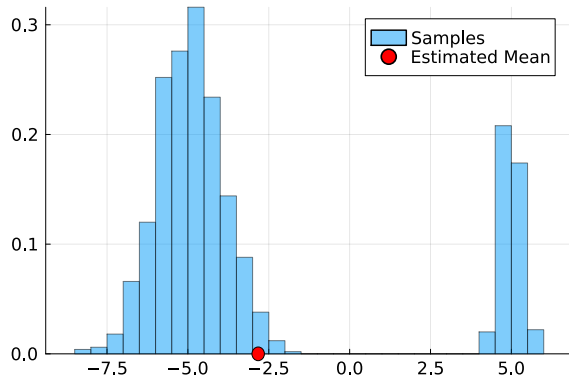


Unit 6 – Approximate Inference:
Variational Inference

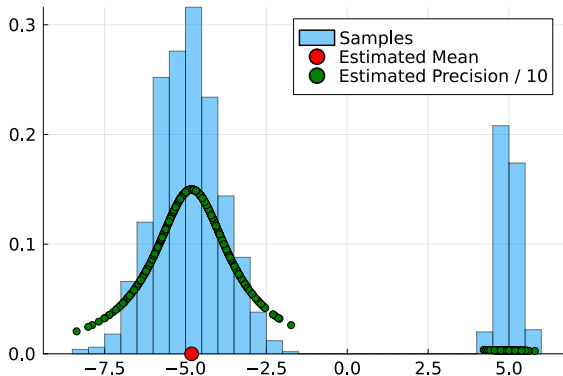
Robust Bayesian Regression: Warm-Up

- **Simple Model:** We have no features so W is a single number T (the average of the observations y_i)
- **Message Schedule:**
 1. Send exact prior messages to T and all S_i
 2. For each observation y_i , update variational message to S_i
 3. For each observation y_i , update variational message to T

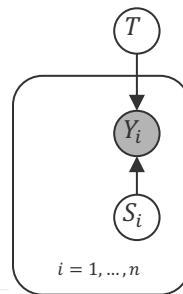
Fixed noise ($\beta^2 = 0.01$)



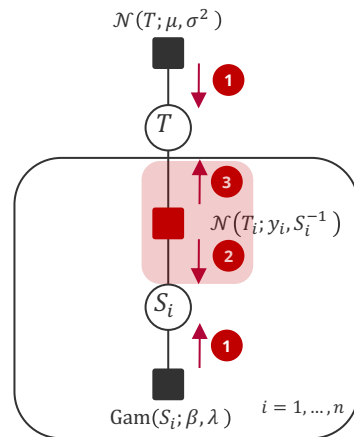
Variable noise



Bayesian Network



Factor Graph



Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

$n = 1000$

$\mu = 0$

$\sigma^2 = 1$

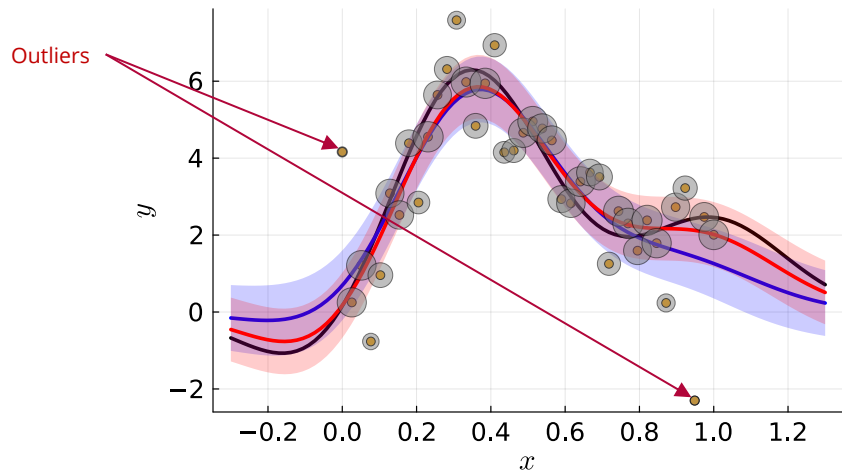
$\beta = 0$

$\lambda = 1$

Robust Bayesian Regression

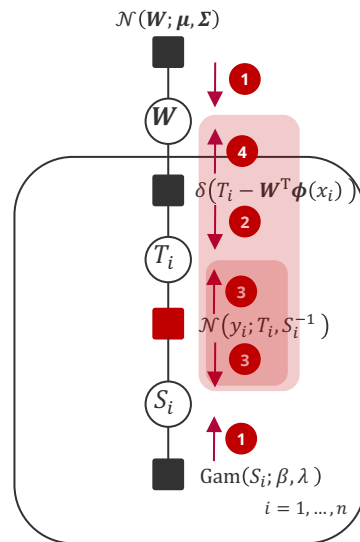
■ Message Schedule:

1. Send exact prior messages to $\mathbf{W}$ and all S_i
- Outer loop iterating over the observations (x_i, y_i) :
 2. Send the message from the projection factor to T_i
 3. Iterate the variational messages from the Gaussian factor to S_i and T_i
 4. Send the message from the projection factor to $\mathbf{W}$



$$\begin{aligned} \mu &= \mathbf{0} \\ \Sigma &= \mathbf{I} \\ \phi(x) &= \begin{bmatrix} \mathcal{N}(x; 0, 0.2^2) \\ \vdots \\ \mathcal{N}(x; 1, 0.2^2) \end{bmatrix} \\ \beta &= 0.5 \\ \lambda &= 1 \\ n &= 40 \end{aligned}$$

Factor Graph



Unit 6 – Approximate Inference:
Variational Inference

Robust Bayesian Ranking

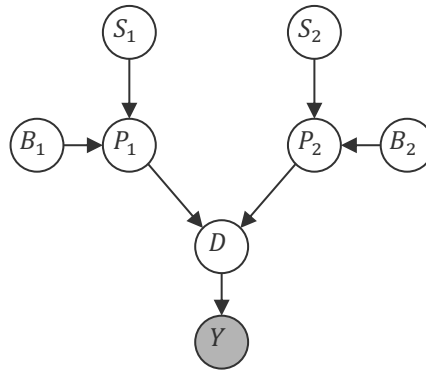
■ Classic:

- Two players have unknown skills S_1 and S_2 with Gaussian belief
- Performance P_i in a match is Gaussian with mean S_i
- Match-outcome $y \in \{-1, +1\}$ means that $y \cdot (P_1 - P_2) > 0$

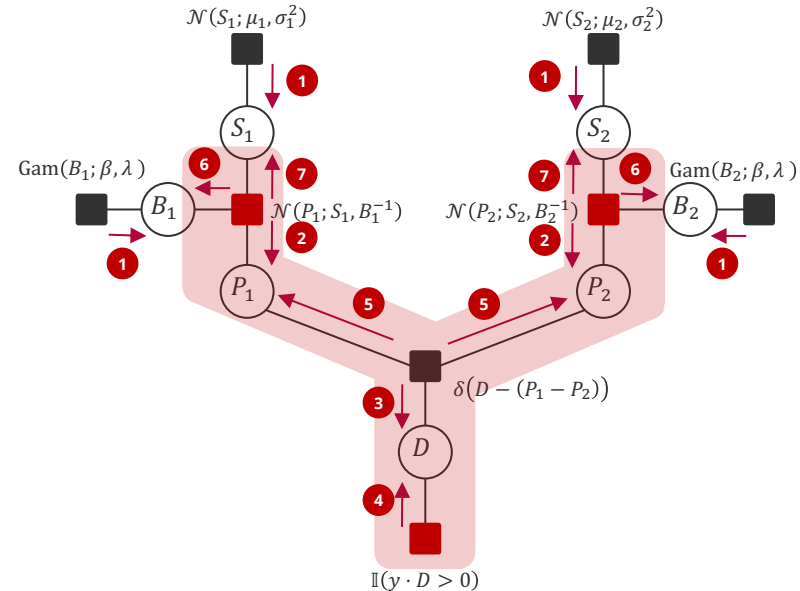
■ Robust:

- We assume an unknown performance variation B_i with Gamma distribution
- Performance P_i in a match is Gaussian with mean S_i and variance B_i^{-1}
- The rest is the same!

Bayesian Network



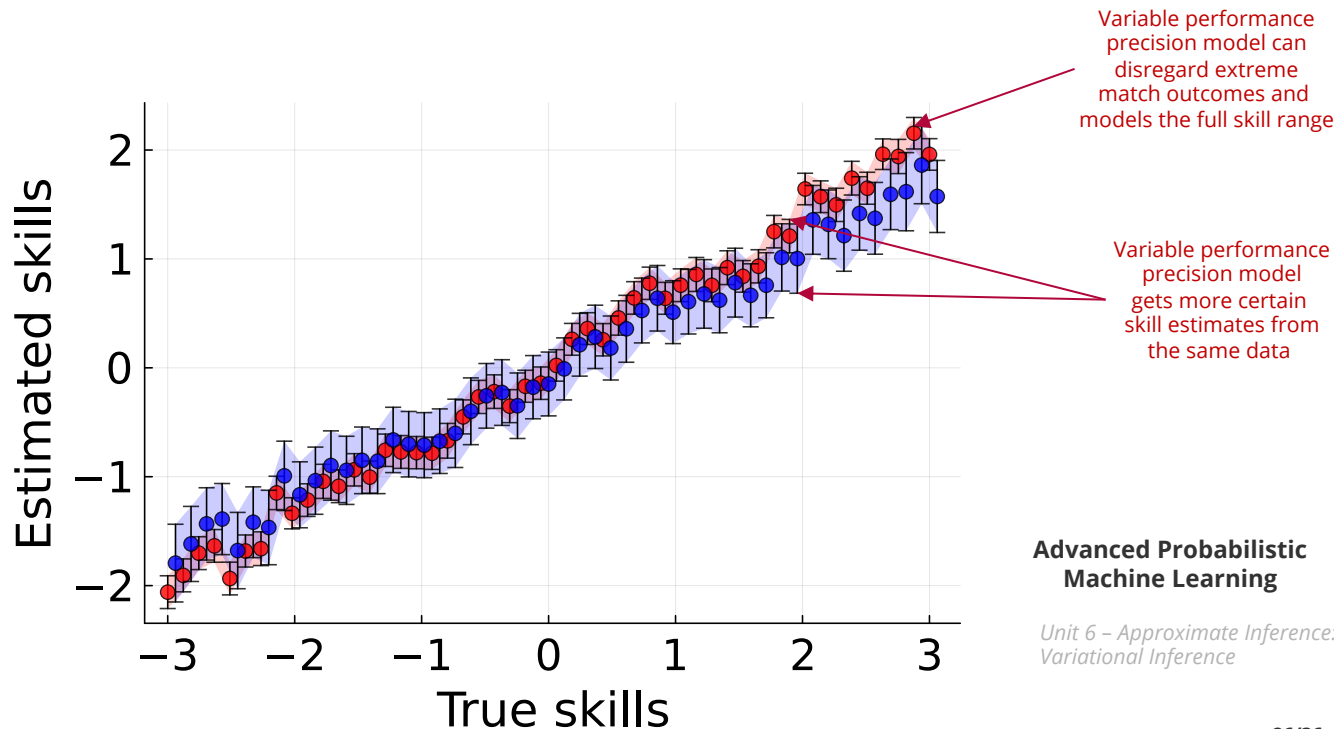
Factor Graph



Robust Bayesian Ranking: Example

■ Experimental Setup:

- We have 50 players with skills evenly distributed in $[-3,3]$
- The prior of the performance precision is $B_i \sim \text{Gam}(0,1)$
- We generate 10,000 matches with random player indices and outcomes using random performance precisions
- We ran TrueSkill Through Time both with fixed (blue) and variable (red) performance precision



Advanced Probabilistic Machine Learning

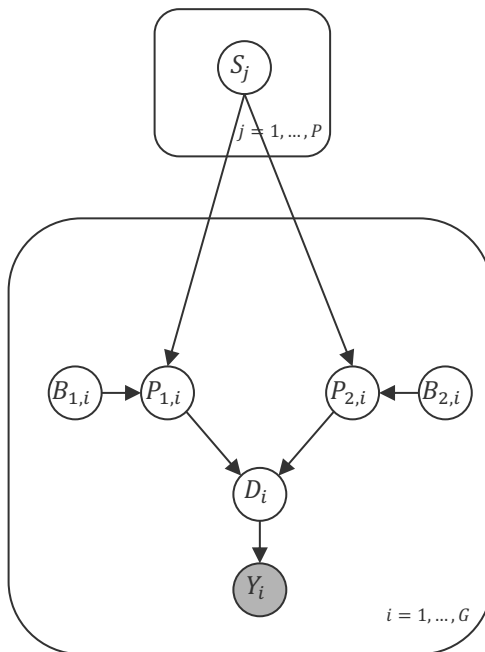
Unit 6 – Approximate Inference: Variational Inference

Robust Ranking: Modelling Performance Precisions

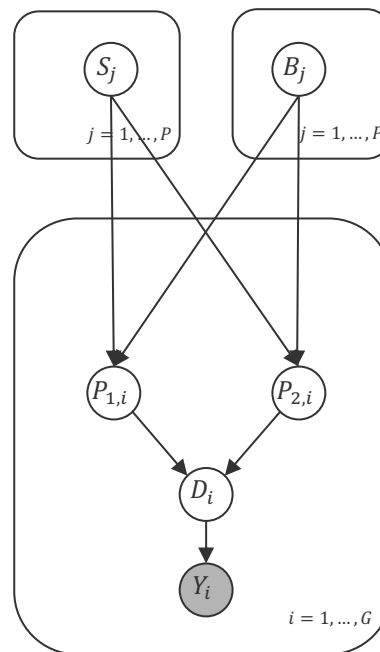
■ Per-Match Precisions:

- Number of skills variables equals number P of players
- Number of performance precisions equals number G of games

Per-Match Performance Precisions



Per-Player Performance Precisions



■ Per-Player Precisions:

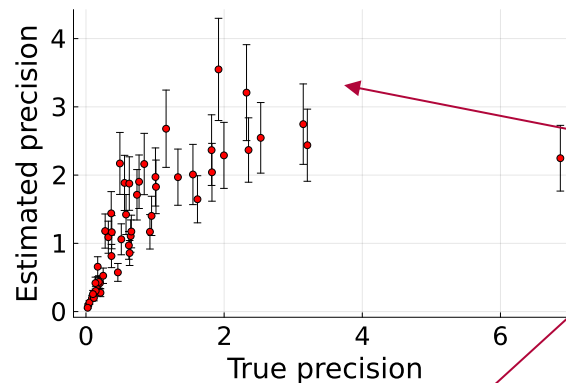
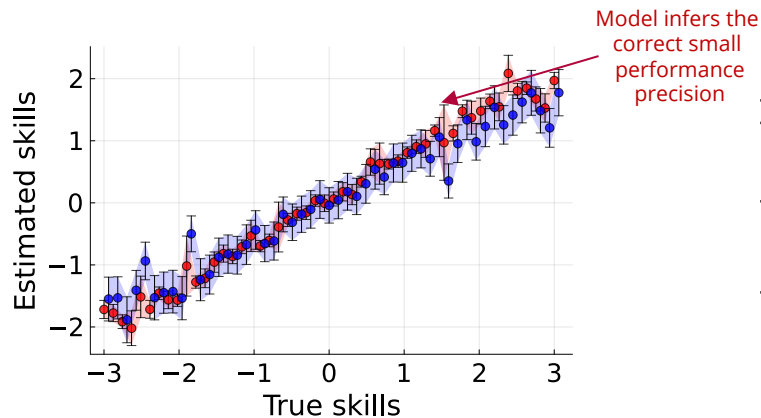
- Number of skills variables equals number P of players
- Number of performance precisions equals number P of players!
- **Pro:** Per-player precisions model a property of players
- **Con:** Loopy graph leads to over-confidence

**Advanced Probabilistic
Machine Learning**

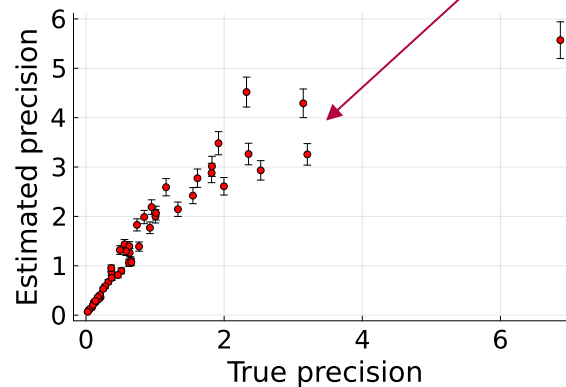
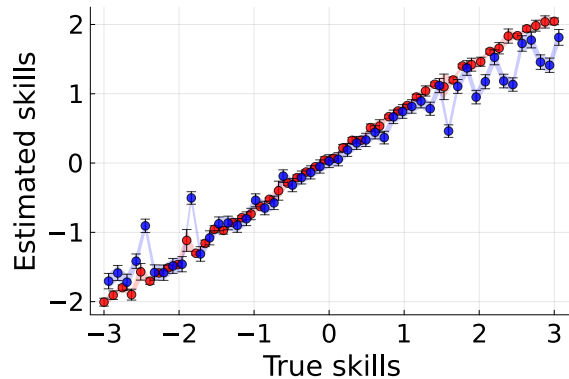
*Unit 6 – Approximate Inference:
Variational Inference*

Robust Bayesian Ranking: Per-Player Precision Example

$G = 10,000$



$G = 100,000$



Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference: Variational Inference

Overview

1. Minimizing Reverse KL-Divergence via ELBO
2. Variational Inference for Gaussian Factors
3. Application: Robust Bayesian Regression and Ranking
4. **Mixture Models**

**Advanced Probabilistic
Machine Learning**

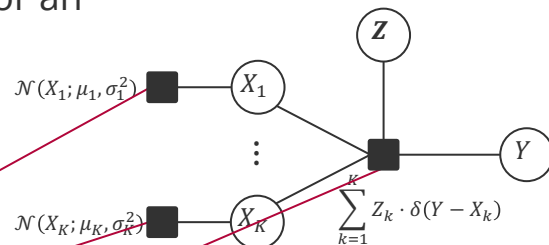
*Unit 6 – Approximate Inference:
Variational Inference*

From Selection Models to Mixture Models

- **Selection Model:** If we have K possible Gaussian causes $X_1, \dots, X_K$ for an observed effect Y and a *categorical* selector $\mathbf{Z} \in \{0,1\}^K$ then

$$f(X_1, \dots, X_K, \mathbf{Z}, Y) = \sum_{k=1}^K \boxed{Z_k} \cdot \delta(Y - X_k)$$

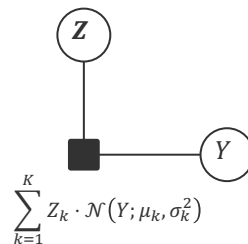
explicit selector



- **Challenge:** What is the marginal distribution if we marginalize out the X_k ?

$$\begin{aligned} g(\mathbf{Z}, Y) &= \int \dots \int f(X_1, \dots, X_K, \mathbf{Z}, Y) dX_1 \dots dX_K \\ &= \int \dots \int \left[\prod_{k=1}^K \mathcal{N}(X_k; \mu_k, \sigma_k^2) \right] \cdot \left[\sum_{k=1}^K Z_k \cdot \delta(Y - X_k) \right] dX_1 \dots dX_K \\ &= \sum_{k=1}^K Z_k \cdot \int \mathcal{N}(X_k; \mu_k, \sigma_k^2) \cdot \delta(Y - X_k) dX_k \\ &= \sum_{k=1}^K Z_k \cdot \mathcal{N}(Y; \mu_k, \sigma_k^2) \end{aligned}$$

Mixture Model

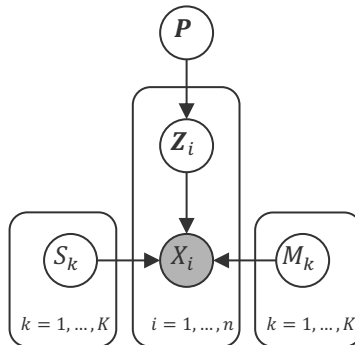


Mixture of Gaussians Model

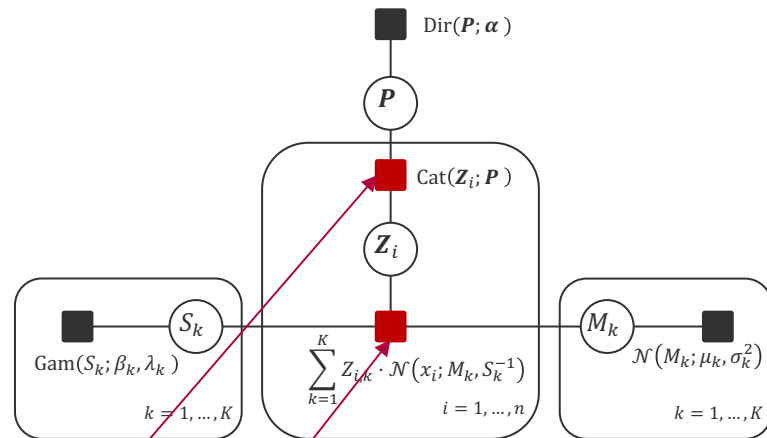
■ Model:

- We assume that there are K Normal distributions with unknown mean M_k and precisions S_k that generate the data $x_i \in \mathbb{R}$
- We assume that each Normal distribution is selected for generating the data with unknown probability P_k
- For each data item, we sample the index j with probability P_j and then sample x_i from $\mathcal{N}(\cdot; M_j, S_j^{-1})$
- We have a Dirichlet prior on $\mathbf{P}$, a Gamma prior on each S_k and a Normal prior on each M_k

Bayesian Network



Factor Graph



Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

We do not know the approximate messages for these two types of factors yet!

Dirichlet-Categorical Factor

$$f(\mathbf{Z}, \mathbf{P}) = \text{Cat}(\mathbf{Z}; \mathbf{P}) = \prod_{k=1}^K P_k^{Z_k}$$

$$= \exp\left(\sum_{k=1}^K Z_k \cdot \log(P_k)\right)$$

$$\propto \exp\left(\begin{bmatrix} Z_1 \\ \vdots \\ Z_K \end{bmatrix}^T \begin{bmatrix} \log(P_1) \\ \vdots \\ \log(P_K) \end{bmatrix}\right)$$

$\mathbf{o}_P(\mathbf{Z})$ $\mathbf{T}_P(\mathbf{P})$

$$\hat{m}_{f \rightarrow P}(\mathbf{p}) \propto \exp\left(E_{\mathbf{Z} \sim \hat{p}_Z}[\mathbf{o}_P(\mathbf{Z})]^T \mathbf{T}_P(\mathbf{p})\right)$$

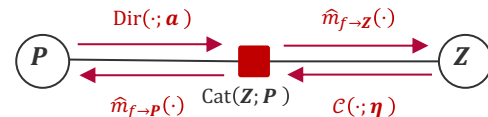
$$\propto \text{Dir}(\mathbf{p}; E_{\mathbf{Z} \sim \hat{p}_Z}[\mathbf{Z}] + \mathbf{1})$$

$$\propto \exp\left(\begin{bmatrix} \log(P_1) \\ \vdots \\ \log(P_K) \end{bmatrix}^T \begin{bmatrix} Z_1 \\ \vdots \\ Z_K \end{bmatrix}\right)$$

$\mathbf{o}_Z(\mathbf{P})$ $\mathbf{T}_Z(\mathbf{Z})$

$$\hat{m}_{f \rightarrow Z}(\mathbf{z}) \propto \exp\left(E_{\mathbf{P} \sim \hat{p}_P}[\mathbf{o}_Z(\mathbf{P})]^T \mathbf{T}_Z(\mathbf{z})\right)$$

$$\propto \mathcal{C}\left(\mathbf{z}; E_{\mathbf{P} \sim \hat{p}_P} \begin{bmatrix} \log(P_1) \\ \vdots \\ \log(P_K) \end{bmatrix}\right)$$



$$\hat{p}_Z(\mathbf{z}) = \mathcal{C}(\mathbf{z}; \boldsymbol{\eta}) \cdot \hat{m}_{f \rightarrow Z}(\mathbf{z})$$

$$\hat{p}_P(\mathbf{p}) = \text{Dir}(\mathbf{p}; \boldsymbol{\alpha}) \cdot \hat{m}_{f \rightarrow P}(\mathbf{p})$$

If $\mathbf{P} \sim \text{Dir}(\cdot; \boldsymbol{\alpha})$ then
 $E[\log(P_i)] = \psi(\alpha_i) - \psi(\boldsymbol{\alpha}^T \mathbf{1})$

**Advanced Probabilistic
Machine Learning**

Unit 6 – Approximate Inference:
Variational Inference

- The two messages are coupled due to the expectation of $\mathbf{P}$ and $\mathbf{Z}$!
- There is no analytical solution to solve them simultaneously
- Thus, they get iterated until convergence from a start-point given by the incoming messages $\text{Dir}(\cdot; \boldsymbol{\alpha})$ and $\mathcal{C}(\cdot; \boldsymbol{\eta})$

Mixture Factor for 1D Gaussians

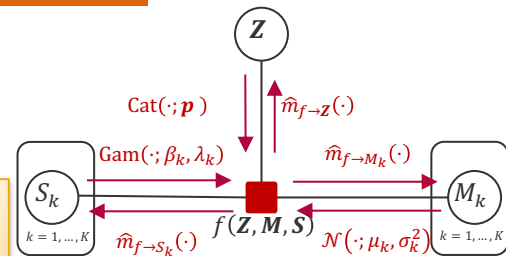
$$f(\mathbf{Z}, \mathbf{M}, \mathbf{S}) = \prod_{k=1}^K [\mathcal{N}(x; M_k, S_k^{-1})]^{Z_k} = \exp \left(\sum_{k=1}^K Z_k \cdot \frac{1}{2} [\log(S_k) - \log(2\pi) - S_k \cdot (x - M_k)^2] \right)$$

$$\begin{aligned} & \propto \exp \left(\frac{1}{2} \cdot \begin{bmatrix} Z_k \\ Z_k(x - M_k)^2 \end{bmatrix}^T \begin{bmatrix} \log(S_k) \\ -S_k \end{bmatrix} \right) \\ & \propto \exp \left(\begin{bmatrix} x \cdot Z_k \cdot S_k \\ Z_k \cdot S_k \end{bmatrix}^T \begin{bmatrix} M_k \\ -\frac{M_k^2}{2} \end{bmatrix} \right) \\ & \propto \exp \left(\frac{1}{2} \cdot \begin{bmatrix} \log(S_1) - S_1 \cdot (x - M_1)^2 \\ \vdots \\ \log(S_K) - S_K \cdot (x - M_K)^2 \end{bmatrix}^T \begin{bmatrix} Z_1 \\ \vdots \\ Z_K \end{bmatrix} \right) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow S_k}(s_k) & \propto \exp \left(E_{\mathbf{Z} \sim \hat{p}_{\mathbf{Z}}, \mathbf{M} \sim \hat{p}_{\mathbf{M}}} [\mathbf{O}_{S_k}(\mathbf{Z}, \mathbf{M})]^T \mathbf{T}_{S_k}(s_k) \right) \\ & \propto \text{Gam} \left(s_k; \frac{E[Z_k]}{2}, \frac{E[Z_k] \cdot [x^2 - 2x \cdot E[M_k] + E[M_k^2]]}{2} \right) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow M_k}(m_k) & \propto \exp \left(E_{\mathbf{Z} \sim \hat{p}_{\mathbf{Z}}, \mathbf{S} \sim \hat{p}_{\mathbf{S}}} [\mathbf{O}_{M_k}(\mathbf{Z}, \mathbf{S})]^T \mathbf{T}_{M_k}(m_k) \right) \\ & \propto \mathcal{G}(m_k; x \cdot E[Z_k] \cdot E[S_k], E[Z_k] \cdot E[S_k]) \end{aligned}$$

$$\begin{aligned} \hat{m}_{f \rightarrow \mathbf{Z}}(\mathbf{z}) & \propto \exp \left(E_{\mathbf{M} \sim \hat{p}_{\mathbf{M}}, \mathbf{S} \sim \hat{p}_{\mathbf{S}}} [\mathbf{O}_{\mathbf{Z}}(\mathbf{S}, \mathbf{M})]^T \mathbf{T}_{\mathbf{Z}}(\mathbf{z}) \right) \\ & \propto \mathcal{C} \left(\mathbf{z}; \begin{bmatrix} E[\log(S_1)] - E[S_1] \cdot (x^2 - 2x \cdot E[M_1] + E[M_1^2])^2 \\ \vdots \\ E[\log(S_K)] - E[S_K] \cdot (x^2 - 2x \cdot E[M_K] + E[M_K^2])^2 \end{bmatrix} \right) \end{aligned}$$



$$\hat{p}_{\mathbf{Z}}(\mathbf{z}) = \text{Cat}(\mathbf{z}; \mathbf{p}) \cdot \hat{m}_{f \rightarrow \mathbf{Z}}(\mathbf{z})$$

$$\hat{p}_{\mathbf{M}}(\mathbf{m}) = \prod_{k=1}^K \mathcal{N}(m_k; \mu_k, \sigma_k^2) \cdot \hat{m}_{f \rightarrow M_k}(m_k)$$

$$\hat{p}_{\mathbf{S}}(\mathbf{s}) = \prod_{k=1}^K \text{Gam}(s_k; \beta_k, \lambda_k) \cdot \hat{m}_{f \rightarrow S_k}(s_k)$$

Advanced Probabilistic Machine Learning

Unit 6 – Approximate Inference:
Variational Inference

- The $2K + 1$ messages are coupled due to the expectation of $\mathbf{Z}$, $\mathbf{M}$ and $\mathbf{S}$
- Iterate to solve for the parameters of the messages $\hat{m}_{f \rightarrow \mathbf{Z}}(\cdot)$, $\hat{m}_{f \rightarrow M_k}(\cdot)$ and $\hat{m}_{f \rightarrow S_k}(\cdot)$

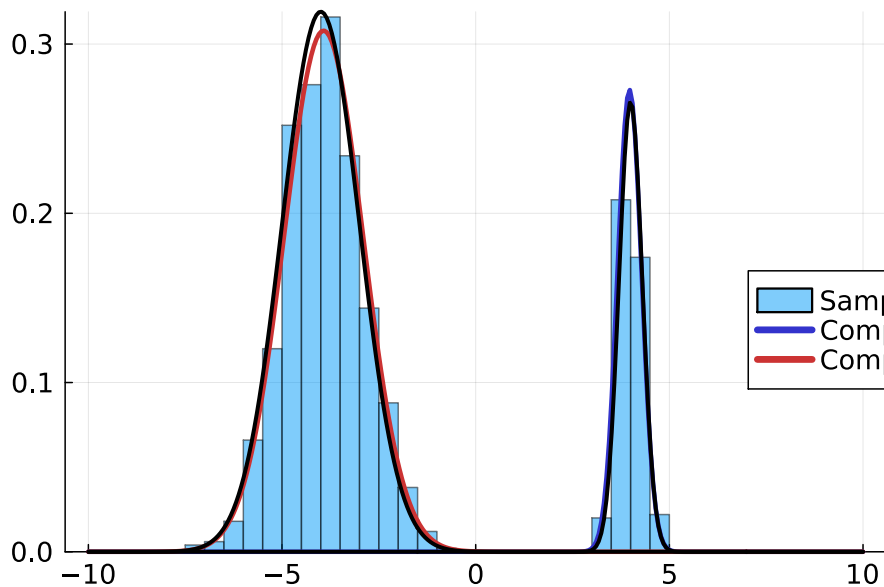
Mixture of 1D Gaussians: Example

$n = 1000$

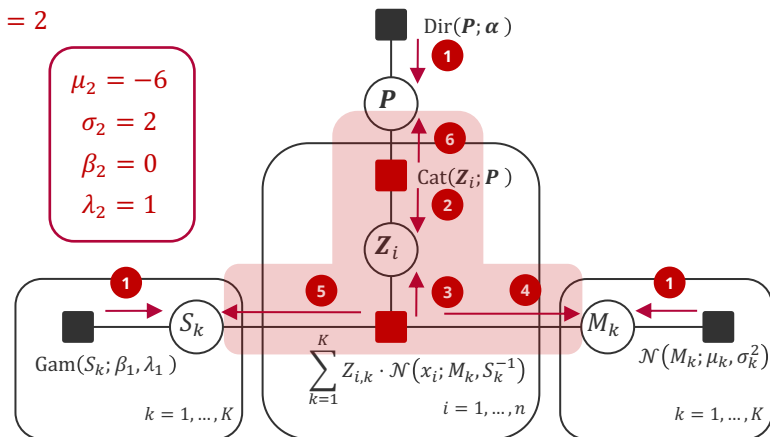
$K = 2$

$\mu_1 = 6$
 $\sigma_1 = 2$
 $\beta_1 = 0$
 $\lambda_1 = 1$

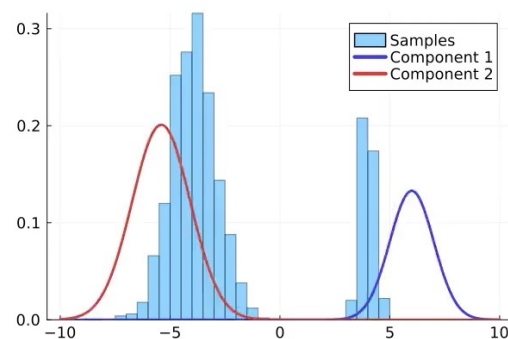
$\mu_2 = -6$
 $\sigma_2 = 2$
 $\beta_2 = 0$
 $\lambda_2 = 1$



Samples
Component 1
Component 2



One frame per training data messages



Samples
Component 1
Component 2

■ Minimizing Reverse KL-Divergence via ELBO

- Minimizing the KL-divergence on the marginal can generate invalid message
- Minimize the *reverse* KL divergence but there is no closed-form solution for the minimizer
- The solution is obtained by iterating fixed-point equations until convergence

■ Variational Inference for Gaussian Factors

- One of the most prominent examples where variational message passing works if the Gaussian density when the precision (i.e., inverse variance) has a Gamma distribution
- Variational message passing leads to an underestimate of the variance!

■ Robust Bayesian Regression and Ranking

- Bayesian regression with unknown precision *per observation* becomes robust to outliers
- TrueSkill can also be extended with variable performance precision *per match* or *per player*. These algorithms are more robust (and have not yet been published!)

■ Mixture Models

- Selection of possible causes for an observation always leads to mixture factors
- For all mixture components in the exponential family, variational inference can be used!

See you next week!